



# Fabrication of solution-processed light-emitting devices from a single deposition step

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## ABSTRACT

Simple, scale-up, and low-cost processing is required for emerging electronic/optoelectronic applications, such as smart packaging, devices for instant information access, large-area billboards, and many others. Over the past decades, different types of new electronic materials and devices have been developed, aiming at the fabrication of thin-film light-emitting devices with flexible and lightweight capabilities. Light-emitting device structures that avoid using indium–tin oxide (ITO) as transparent electrodes are highly desirable, representing a considerable reduction in fabrication costs and an alternative to future material (indium) shortage. Despite the use (or not) of ITO, even the simplest device structure needs to undergo at least three deposition steps, comprising a transparent electrode (also used in many cases as the substrate), the electroluminescent active layer, and a back electrode. Here, we present a proof of concept for the fabrication of electroluminescent devices, with the primary innovation being the significant reduction in processing steps. We demonstrate herein a single-step process where the electroluminescent active layer and the transparent electrode are formed simultaneously by drop casting a multiphasic suspension onto a metallic foil, which acts simultaneously as the device back electrode and substrate. The resulting device exhibited a turn-on voltage of 23 V, a luminance on the order of 10 cd/m<sup>2</sup>, and a luminous efficacy on the order of 10<sup>−3</sup> cd/A. Although these performance parameters are lower compared to more mature technologies, such as polymer light-emitting diodes (PLEDs) and polymer light-emitting electrochemical cells (PLECs), our findings introduce a novel fabrication concept for electroluminescent devices, broadening the potential for the development of emerging technologies.

## 1 Introduction

The demand for new types and structures of light-emitting devices has been continuously increasing in the latest decades, mainly because of the ubiquitous

presence of screens and displays, from TV sets and monitors to laptops, tablets, smartphones, smartwatches, and other personal gadgets, as well as the unprecedented need visual signaling in emerging applications like automotive interiors and smart

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packing. Indoor illumination also has been transformed drastically since the development of full color, bright light-emitting diodes [1], unfolding an enormous variety of possibilities and applications [2].

The development of active matrix (AM) displays based on liquid crystals, LCDs [3] and on organic light-emitting diodes, OLEDs [4, 5], enabled the almost complete replacement of cathode ray tubes (CRT) based displays and expanded the possibilities of new applications, especially for mobile devices (smartphones, tablets, and smartwatches). Recent advances in inorganic light-emitting diodes (LEDs) technology such as higher emission efficiency and emission at shorter wavelengths (in the UV and blue ranges of the visible spectrum) permitted the emission of bright white light, finding immediate applications for the most varied purposes, including backlights for AMLCDs and lamps for indoor illumination.

Emerging applications, such as disposable light-emitting indicators for smart packaging and wearable health monitoring devices [6, 7], have created an increasing pursuit for new architectures and processing methods. OLEDs, polymer light-emitting diodes, PLEDs [8], quantum-dot light-emitting diodes, QLEDs [9], polymer light-emitting electrochemical cells, PLECs [10], and electroluminescent devices produced with light-emitting composites, LECELs [11, 12] are technologies based on thin films which are compatible with large area and low-cost deposition. The study of the electrical and optical properties of such light-emitting devices opens up the possibility of obtaining products with new designs, such as ultra-thin and flexible light-emitting devices [13], as well as high-performance ultra-thin displays [14].

Although OLEDs and QLEDs are currently available commercial technologies, being mainly used in devices, such as TV sets and smartphone displays, the production costs and manufacturing scale-up of these technologies are limited by their complex fabrication process, comprising multiple layers, which are commonly deposited by thermal evaporation, RF-sputtering or other high-vacuum physical vapor deposition techniques. The fabrication process usually requires at least three layers: (i) a transparent anode (ITO), (ii) a low-work function cathode, and (iii) the active emitting layer [8, 9, 15]. In most of the applications, more complex architectures are used, introducing electron/hole injecting/conducting layers and additional thin or ultra-thin interlayers, to achieve higher overall performance [13, 14]. This set of characteristics leads

to processing restrictions such as manufacturing in a clean room and/or inert atmosphere conditions, complicating the fabrication of low-cost and disposable devices. On the other hand, PLECs and LECELs are solution-processed devices that have no major limitations regarding film thickness and electrode work function, dispensing the need for multiple, thin-layer deposition [16]. PLECs are, however, extremely sensitive to moisture and oxygen, requiring inert atmosphere conditions, in contrast to LECELs, which can be produced in ambient conditions, providing a robust, simple, and low-cost technology to produce electroluminescent devices with large-area capabilities.

LECEL active materials are hybrid composites which combine the electrical conductivity of doped conjugated polymers, such as polyaniline (PANI) and derivatives or poly(3, 4-ethylenedioxythiophene):poly(styrene sulfonate) (PEDOT:PSS) and the light-emitting characteristics of inorganic luminescent powders, such as  $\text{Zn}_2\text{SiO}_4\text{:Mn}$  [17, 18]. This type of light-emitting material can be processed at ambient conditions because LECELs employ relatively thick ( $\sim 100\text{ }\mu\text{m}$ ) active layers and use high work function electrodes. However, in previous reports, an RF-sputtered ITO bottom electrode and a thermally evaporated metallic top electrode were still necessary for device proper operation. We introduce herein the concept of a solution-processed light-emitting device produced in a single deposition step method, discarding the need for electrode deposition and other multilayer deposition steps.

## 2 Materials and methods

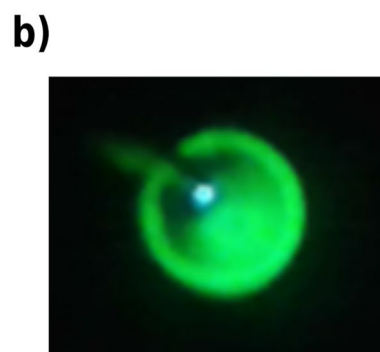
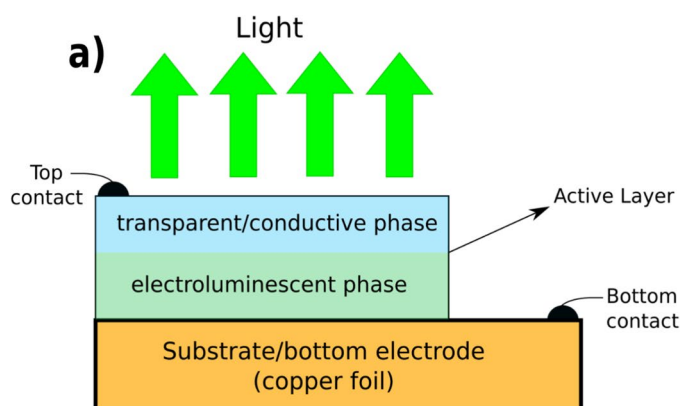
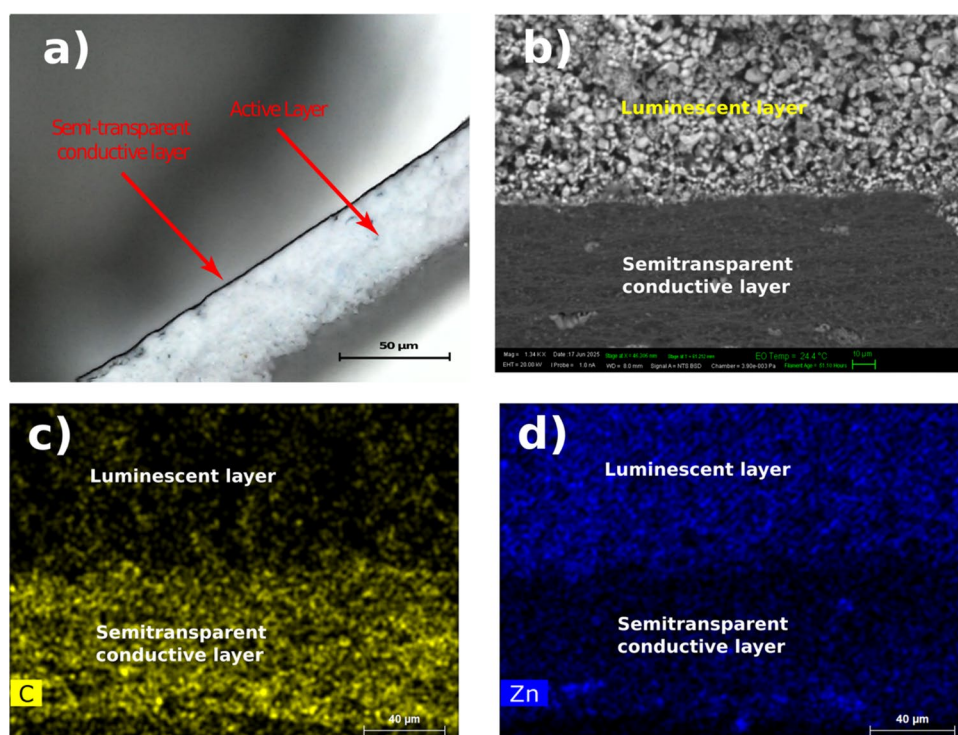
The device active layer used in our experiments comprised hybrid composites of Mn-doped zinc silicate ( $\text{Zn}_2\text{SiO}_4\text{:Mn}$ , P1-GK from Phosphor Technology) and PEDOT:PSS (purchased from Sigma-Aldrich). The suspension was produced by dispersing the  $\text{Zn}_2\text{SiO}_4\text{:Mn}$  micro-particles into a PEDOT:PSS water-based solution and silica-organic material (3-glycidyloxypropyl)trimethoxysilane (GPTMS, Sigma-Aldrich). The single-step deposition method is based on drop casting the non-stable multiphasic suspension onto copper foils, which act as both substrate and bottom electrode. After the deposition of the unstable suspension, the inorganic micro-particles slowly decanted toward the metallic substrate, spontaneously creating a biphasic structure which can be considered as two different wet

layers: a semi-transparent conductive layer on top, composed mainly by the PEDOT:PSS/GPTMS phase and an electroluminescent layer composed mainly by  $\text{Zn}_2\text{SiO}_4\text{:Mn}$  micro-particles, intermediated by the PEDOT:PSS/GPTMS mixture, as shown in Fig. 1. The device in its final structure was obtained by spontaneous evaporation of the solvent and further baking at low temperatures (about 120 °C). Since the top layer is conductive and semi-transparent, there is no need to use a transparent electrode layer like indium–tin oxide

(ITO) or similar, and it is used as the device top electrode. The device architecture is represented in Fig. 2a, whereas Fig. 2b shows a photograph of a device operating in a dark room.

Figure 1a shows an optical microscopy image from the cross-section of a fractured sample. In Fig. 1b, one can observe, from the scanning electron microscope (SEM) image of the cross section, the phase separation in the device active layer. The device top phase is clearly more compact (Fig. 2b) than the bottom

**Fig. 1** Cross-sectional images of the sample: **a** optical microscopy image, **b** scanning electron microscopy (SEM) image, **c** energy-dispersive X-ray spectroscopy (EDS) elemental map for carbon, and **d** EDS elemental map for zinc. All images were obtained from the cross-sections of different samples fractured in liquid nitrogen. The samples used for SEM and EDS analyses are significantly thicker than those employed in device fabrication, as illustrated in the optical microscopy image



**Fig. 2** **a** Representation of the architecture of the single-step processed LECEL device (not-to-scale drawing) and **b** single-step processed LECEL operating in a dark room

phase, which is consistent with the picture that the  $\text{Zn}_2\text{SiO}_4\text{:Mn}$  decantates, forming a more porous phase in the bottom layer and leaving a smoother phase in the top layer, composed mainly by PEDOT:PSS. The energy-dispersive X-ray spectroscopy (EDS) mapping for C (Fig. 1c) and for Zn (Fig. 1d) serves to corroborate that the top layer is more dense and richer in C than the bottom layer (Fig. 2c), whereas the bottom layer is richer in Zn (from the zinc silicate).

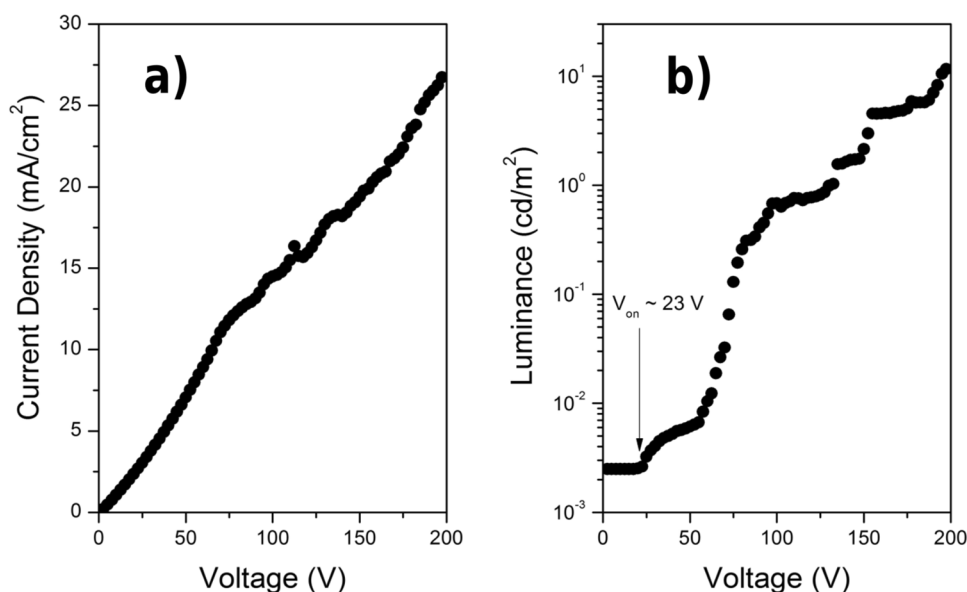
### 3 Results and discussion

The electrical characteristics of the devices were determined by d.c. current–voltage ( $I$ – $V$ ) measurements using a Keithley 2410 source meter unit (SMU), with the electroluminescence (EL) being simultaneously measured using a calibrated photodiode connected to a Keithley 6517B electrometer, resulting in the luminance vs. voltage ( $L$ – $V$ ) curves. In the experiments, the forward bias condition was defined considering the metallic bottom electrode as the cathode and the semi-transparent conducting top layer as the anode. Figure 3a depicts the device current density as a function of the applied voltage at forward bias conditions, presenting an approximately linear behavior. In this condition, no significant potential barriers for charge injection from electrodes are expected, as a consequence of the appropriate energy matching at both electrode interfaces. For the top electrode, the active

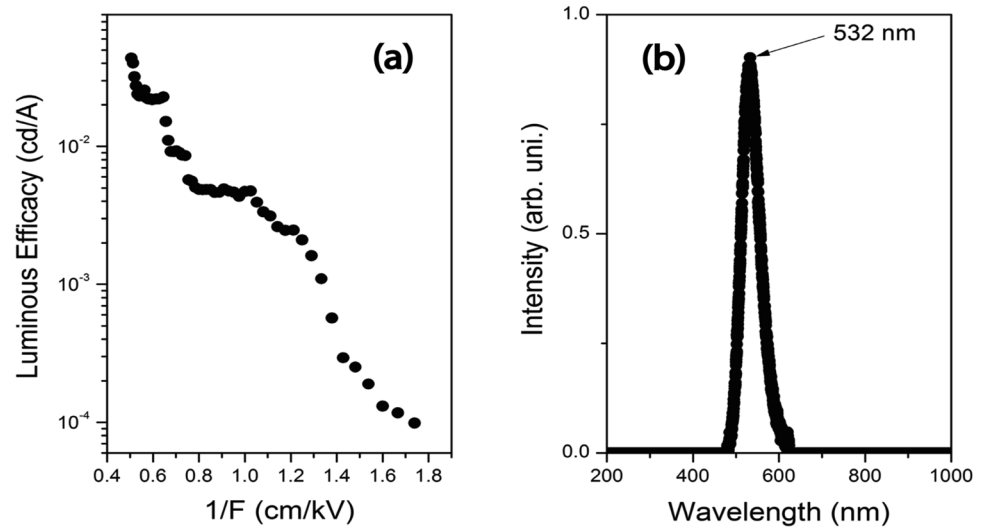
material and the semi-transparent electrode are both composed of PEDOT:PSS, resulting in a negligible energy barrier for charge injection. At the bottom electrode interface, the work function of the copper electrode ( $\sim 4.7$  eV) [19] is lower than the HOMO energy for PEDOT:PSS ( $\sim 5.0$  eV) [20, 21], resulting in an almost ohmic contact, as confirmed by the results in Fig. 3a.

The device turn-on voltage ( $V_{\text{on}}$ ) was determined from the luminance vs. applied voltage curve, Fig. 3b, laying at approximately 23 V, which corresponds to an averaged electric field of 0.6 kV/cm. The turn-on voltage and electric field of the single-step processed LECEL were quite similar to those of previously reported LECELs [11, 17]; however, since the device thickness usually varied in the 300–500  $\mu\text{m}$  range, the observed operating voltage for different devices varied accordingly, maintaining nearly the same averaged operating electric field. Profilometer measurements (shown in Fig. 1S) were carried out to determine the device mean roughness of 6.5  $\mu\text{m}$  (for 10 different measurements), which represents less than 2% of the device average thickness. In the applied voltage range shown in Fig. 3b, an exponential increase of more than three orders of magnitude was obtained for the device luminance, instead of the linear dependence observed for current density vs. applied voltage. Therefore, the luminous efficacy has risen almost three orders of magnitude, with a maximum value of 0.04 cd/A for 27  $\text{mA}/\text{cm}^2$ , as shown in Fig. 4a. Although the performance parameters

**Fig. 3** **a** Current density vs. voltage and **b** luminance vs. voltage obtained from a single-step processed LECEL



**Fig. 4** Single-step processed LECEL: **a** luminous efficacy as a function of the inverse electric field and **b** electroluminescence spectrum



achieved in this study are comparable to those reported for LECEL devices, they remain significantly lower than those of more studied and mature technologies, such as polymer light-emitting diodes (PLEDs) and polymer light-emitting electrochemical cells (PLECs), which can exhibit luminance levels exceeding  $10^5$  cd/m<sup>2</sup>, luminous efficacies above 5 cd/A, and turn-on voltages below 10 V [22, 23].

Figure 4b shows the device electroluminescence spectrum, recorded using a portable USB spectrometer (Ocean Optics USB 2000+). The emission spectrum is characterized by a narrow-band spectrum with a 532 nm dominant wavelength. The green emission peak is assigned to the electronic transition  $^4T_1(G) \Rightarrow ^6A_1(S)$  of  $Mn^{2+}$  ions in the  $Zn_2SiO_4:Mn$  lattice [24]. Impact excitation of  $Mn^{2+}$  ions occurs by injection of charge carriers from the conductive matrix into the inorganic micro-particles, which leads to the typical LECEL characteristic shown in Fig. 4a [25]. The relatively weak electroluminescence efficacy ( $\sim 10^{-2}$  cd/A) originates from a low-charge injection current into the inorganic micro-particles insulating phase. Therefore, the electrical current that can be converted into light is much lower than that in percolated conductive channels. This characteristic leads to the linear current density vs. voltage behavior shown in Fig. 3a as well as to a low device efficacy.

## 4 Conclusion

We demonstrated the possibility to produce LECEL devices using a single deposition step method, which does not need the use of transparent substrates as ITO, nor the deposition of a top metallic electrode by thermal evaporation, sparing time and processing materials and processes. The device efficacy can be, in principle, improved by better matching energy levels of electroluminescent micro-particles and conductive matrix electronic states, by controlling conductive matrix conductivity and its weight concentration. In addition, constructive parameters, such as film thickness, need to be controlled to improve performance characteristics, such as turn-on voltage. Therefore, this architecture creates the possibility for future research and development, including investigations aimed at evaluating and enhancing reproducibility, operational lifetime, spectral stability, angular emission characteristics, light extraction efficiency, and other performance parameters. In addition, this new concept of light-emitting device production has a potential impact on the printing research field toward ITO-free light-emitting devices and low-cost electronics.

## Author contributions

All authors contributed to the study's conception and design. Material preparation, data collection, and analysis were carried out by Luiza Gualter Ramires, Danilo dos Santos, and Rafael Jesus Gonçalves Rubira. Giovanni Gozzi and Lucas Fugikawa-Santos wrote the first draft of the manuscript, and all authors provided feedback on previous versions. All authors read and approved the final manuscript.

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## Data availability

The data generated, processed, and collected throughout this study were obtained using specialized equipment and will be properly stored and protected. The data will be analyzed and maintained on hardware, in addition to being preserved on physical media in the laboratory and authorized personal devices. Additionally, the data will be made available on a cloud storage platform linked to the institutional email of the Universidade Estadual Paulista (UNESP).

To ensure transparency and accessibility, the data will also be shared in the UNESP Scientific Data Repository (<https://repositorio.unesp.br/>), in accordance with the institutional policy for the storage and availability of scientific data.

## Declarations

**Conflict of interest** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s10854-025-15692-0>.

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